Students in Research

Georgia Institute of Technology - Group 2 Social / Behavioral Research Investigators and Key Personnel

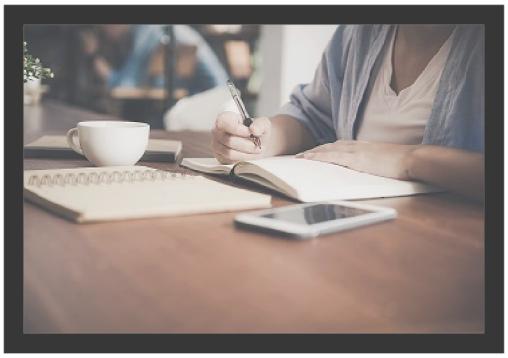
Students in Research

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This module has been revised and adapted from the original module authored by Susan L. Rose and Monica Abakar.

Introduction



As an undergraduate, graduate, or doctoral student, you may have the opportunity to conduct or assist in research projects involving human subjects (also called participants). Such projects help you gain experience with research methodology in your field and understand and apply ethical principles and guidelines for human

subjects research, as established by federal regulations, professional associations, and individual institutions.

This module is designed to address students as researchers and when students are involved in research as participants. It will review the history and status of key research regulations, the Institutional Review Board (IRB) review process, and general best practices when conducting human subjects research.

Institutions often vary in how they apply certain regulations and how their IRB process works. Before preparing an IRB submission or agreeing to serve as a research assistant, you should talk with your faculty advisor and/or your institutional IRB staff to understand local policies and procedures.

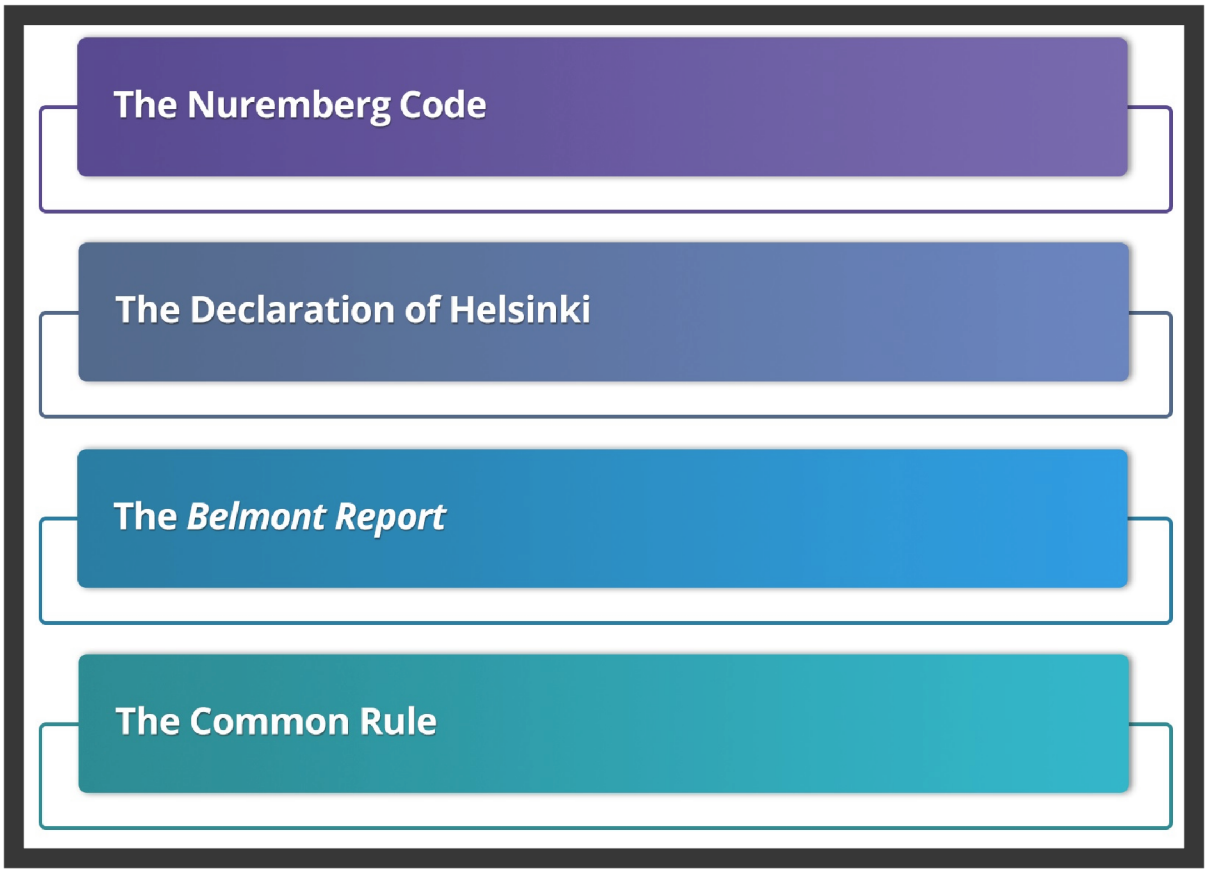
Learning Objectives

By the end of this module, you should be able to:

- Discuss the historical development of regulations associated with protecting human research subjects.
- Identify considerations in defining what constitutes “human subjects research.”
- Describe standard categories of review regarding risks to subjects.
- Review general IRB submission procedures for projects involving human research subjects.
- Discuss strategies for best practices in creating an accurate, robust submission and conducting responsible, ethical research.

History of Ethical Regulations

While several different professional codes of ethics and regulations relate to human subjects research and the protection of human subjects, four primary ones include:



The Nuremberg Code

The Nuremberg Code resulted from the 1940s Nuremberg Military Tribunal, during which Nazi physicians were convicted of unlawful experimentation on concentration camp prisoners (Trials 1949a; Trials 1949b). The Nuremberg Code comprises ten principles regarding the treatment of human participants in research. It establishes the requirement of informed consent and voluntary participation (that is, participants fully understand what the research involves and freely agree to take part). It further states that benefits (whether to society or an individual) must outweigh any potential risks, and that subjects must be able to withdraw at any time without negative consequences.

The Declaration of Helsinki

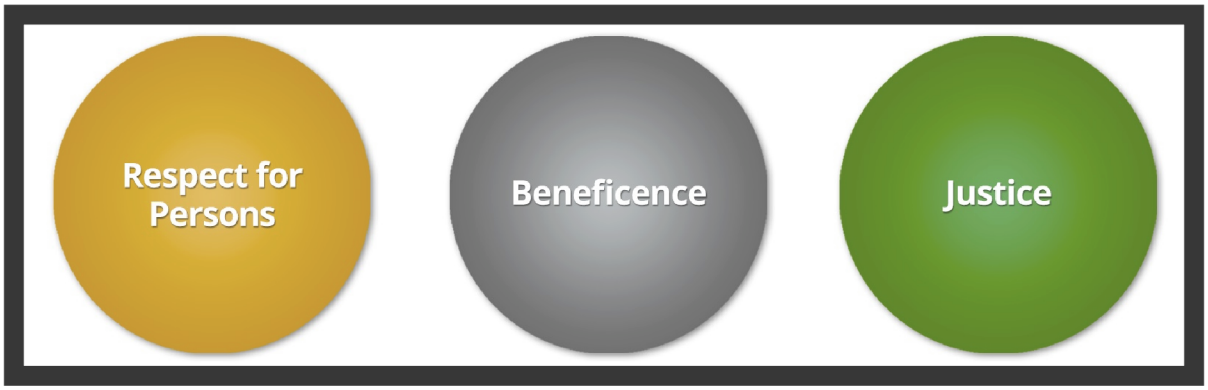
The Declaration of Helsinki (developed by the World Medical Association in 1964 and last revised in 2013) sets forth morally binding ethical principles for medical doctors engaged in biomedical research involving human subjects, as well as identifiable

human biospecimens and data. This declaration, in combination with local and national regulations, helps to establish an international standard of expectations.

The Declaration of Helsinki stresses that only well-trained, qualified individuals should engage in research with human participants, and that physicians must safeguard the health and safety of patients, regardless of broader potential medical benefits. As with the Nuremberg Code, the Declaration of Helsinki emphasizes the importance of informed consent and mitigating harms or risks. Moreover, it notes that some participants or groups may have greater vulnerability and therefore need additional protections. Additionally, the Declaration of Helsinki explicitly states that a research ethics committee (for example, an IRB) must first review and approve all study plans (also referred to as protocols) prior to recruitment and initiation.

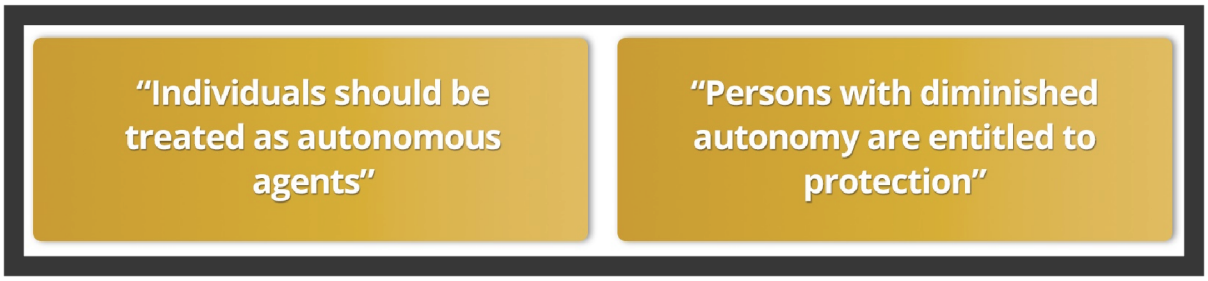
The Belmont Report

The *Belmont Report* (The National Commission 1979) underscores similar ethical considerations for protecting human research subjects. Specifically, the *Belmont Report* identifies three key ethical principles:



Respect for Persons

Respect for persons rests on two ethical convictions:



In other words, you compromise individuals' free will if you do not respect their autonomy (ability to think and act independently). Yet not all individuals may be capable of understanding and voluntarily consenting to research procedures; in these cases (for example, research with young children or people experiencing dementia or other cognitive impairments), extra protections must be put into place, and some potential participants may need to be excluded for their own safety.

Beneficence

Beneficence means making choices that go beyond basic duty and prioritizing participants' well-being throughout a research study. It involves two general rules:



Sometimes research may benefit society at large even though it may not benefit individual participants at the time (for example, discovering future treatments for childhood diseases may justify involving children in research). However, harms to an individual may never exceed benefits to a larger population.

Justice

In general, justice means that equals should be treated equally. In research studies, it means no one should unfairly receive benefits or experience burdens. Historically, some research studies unreasonably excluded women, others inappropriately

recruited marginalized persons of color, and still others collected data from children confined in institutions.

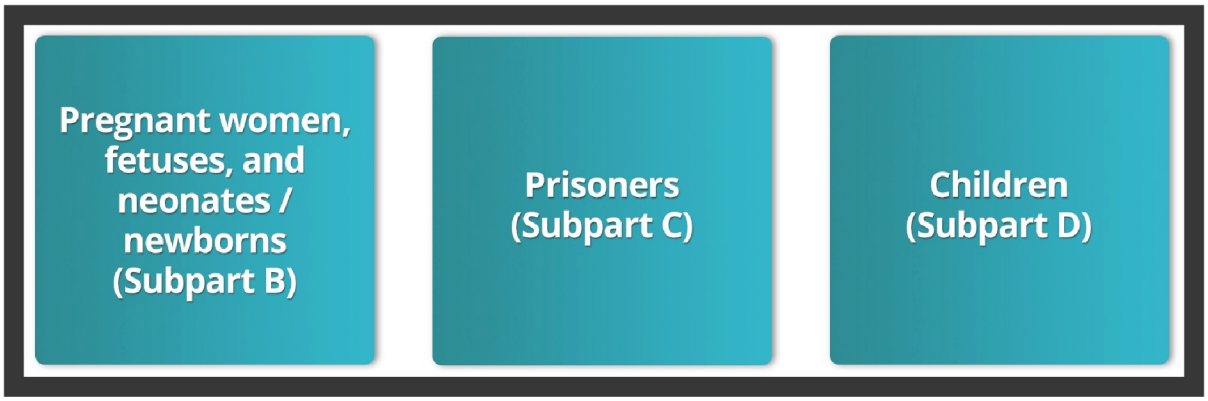
For certain investigations, researchers might legitimately establish inclusion or exclusion criteria surrounding age, race/ethnicity, gender, or economic status.

However, the critical factor regarding justice is that certain individuals (for example those belonging to a marginalized or vulnerable group) may not face greater risks in participation, while those in another group (for example, those in positions of power or with greater wealth) receive benefits from the research. An ethics committee or IRB should evaluate all research studies to ensure the equitable treatment of participants, especially regarding benefits and risks.

The Common Rule

In the United States, researchers often cite the Common Rule, which is the informal name of the U.S. Department of Health and Human Services (HHS) Policy for the Protection of Human Subjects (45 CFR 46, Subpart A) codified in 1991 and revised in 2018.

The Common Rule also offers additional protections for vulnerable groups such as:



Many federal agencies and departments have not adopted subparts B, C, and D. Therefore, only Subpart A is known as the "Common Rule."

Although the rule formally applies only to federally funded research, many academic institutions use the Common Rule as the basis for reviewing studies with human research participants. The Common Rule, together with the documents described above, provides extensive guidance for conducting human subjects research. However, it stops short of issuing rules that apply to every situation and context. As a result, institutions develop additional local policies appropriate for their culture and nature of research.

Learning from Research Abuses and Misconduct

Unfortunately, research abuses and misconduct are not limited to past events (for example the Nazi war crimes and Tuskegee Syphilis Study). Research abuses and misconduct continue due in part to lack of appropriate training or knowledge and intentional disregard for regulations and policies.

For example, in the late 1990s researchers at the University of Pennsylvania Institute for Human Gene Therapy recruited individuals for a clinical trial investigating potential treatment for a rare metabolic disorder (OTCD). The study came to a halt when Jesse Gelsinger—believed to be an otherwise healthy teenager—experienced a fatal reaction to the treatment and died within days of receiving the shot. An investigation of the teen's death revealed multiple ethical lapses in the researchers' practices. They failed to notify the FDA or pause the study when patients experienced serious side effects. In screening potential participants, they ignored that Jesse's pretrial tests showed poor liver function. The researchers also did not tell Jesse about other patients' side effects or the death of two monkeys from the treatment. The Gelsinger family sued the Institute, which was later suspended by the FDA and shut down by the university.

One of the most well-known cases in recent years is the 2014 Facebook Emotional Contagion study. For approximately a week, some individuals' Facebook news feeds were altered to show mostly positive content, whereas others saw primarily negative content. Researchers then studied the "emotional contagion" effect of whether

individuals subsequently posted positive or negative items themselves (Kramer et al. 2014). At the time, the study was not considered human subjects research by the reviewing IRB and data access was consistent with Facebook's data use policy, but it is highly controversial from a research ethics perspective. Many questioned the ethics of conducting a "secret" study that impacted individuals' personal interactions with others, without their informed consent and without appropriate IRB review.

In these cases, ensuring that a study meets ethical and legal standards underscores the principles of respect for persons and voluntary participation. Consulting with an IRB or ethical review board allows researchers at all levels to confirm that they have taken the necessary steps for establishing appropriate participant protections.

What is human subjects research?

The answer to this question may seem simple, but even experienced regulatory professionals often discuss what constitutes human subjects research. Moreover, different institutions may interpret the regulatory definitions in slightly different ways.

Therefore, two questions to consider when you begin thinking about a project are:



Note: This section is based on the Common Rule. State and/or local law (as well as the laws of other countries) may vary.

Does it involve "human subjects?"

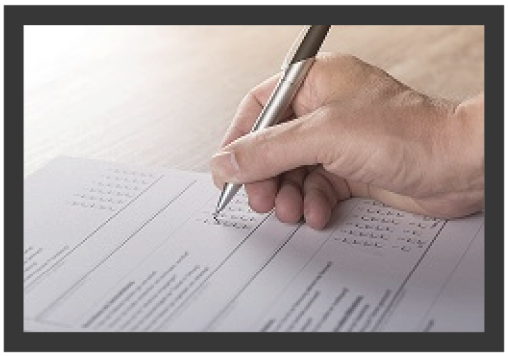


According to the Common Rule, **human subject** refers to "a living individual about whom an investigator (whether professional or student) conducting research: (i) Obtains information or biospecimens through intervention or interaction with the individual, and uses, studies, or analyzes the information or biospecimens; or (ii) Obtains, uses, studies, analyzes, or generates identifiable private information or identifiable biospecimens."

Let us take a moment to unpack the language in this definition. **Note:** We have italicized some terms for emphasis.

- **Living individual** – Data must originate from *live* people; specimens or information from dead people or cadavers do not count.
- **About whom** – Data are *about* living individuals (for example, their beliefs, habits, practices, or blood or saliva specimens). You are gathering information about some*one* not some*thing*.
- **Intervention** – The study involves some procedure or action that affects the individuals or their environment to measure "before and after" results.
- **Interaction** – Communication occurs directly with the participant (for example, in person or by phone, email, videoconference, or online survey)
- **Identifiable private information** – You or another member of the research team can easily determine the participant's identity. Private means that the individual does not reasonably expect observation (in contrast to a public space such as a restaurant or bus), has shared the information for study purposes, and can assume identifiers will not be published without explicit permission (you will learn more about the term "identifiable" in a later section).
- **Biospecimens** – Data taken from a living body such as saliva, urine, blood, tissues, or cells.

Is it "research?"



The Common Rule defines **research** as “a systematic investigation, including research development, testing, and evaluation, designed to develop or contribute to generalizable knowledge.” Additionally, the *Belmont Report* defines research as “an activity designed to test a hypothesis [and] permit conclusions to be drawn” through a formal study plan with a set of procedures to reach an objective.

Activities that federal regulations identify as “not research” include oral history, journalism, biography, literary criticism, legal research and historical scholarship that focuses on specific individuals about whom the information is collected, as well as certain public health surveillance activities under the direction of a public health authority.

However, institutions differ in how they apply these regulatory definitions. For example, although the federal regulations include oral history and student course projects in the “not research” list, some institutions review these activities to ensure the protection of the human subjects providing personal data.

Consider these different scenarios and how they relate to researcher’s intent of “generalizable knowledge.”

If a student wishes to present a capstone paper at a departmental symposium, would that represent "public presentation?"

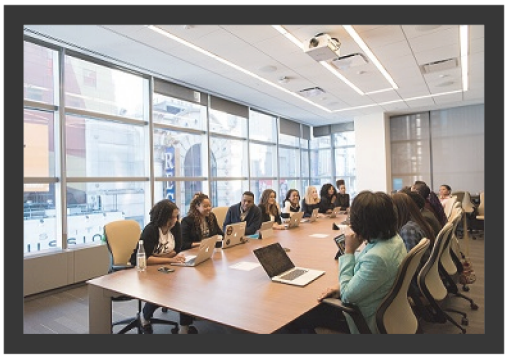
What if a researcher removes all identifying information from data, then later decides to submit a project based on that data for publication?

Clearly distinguishing between studies designed to generalize to a larger audience versus those that intend only to assess operational activities can be challenging. Studies that appear focused on one goal may shift during the project or serve multiple purposes (many studies intended only to improve services, programs, or internal functioning are "quality improvement" or "quality assessment" and do not need IRB approval).

Always check with your IRB early in the process to find out what activities require review and approval. This will help ensure that you correctly follow guidelines and remain in compliance with applicable regulations and policies.

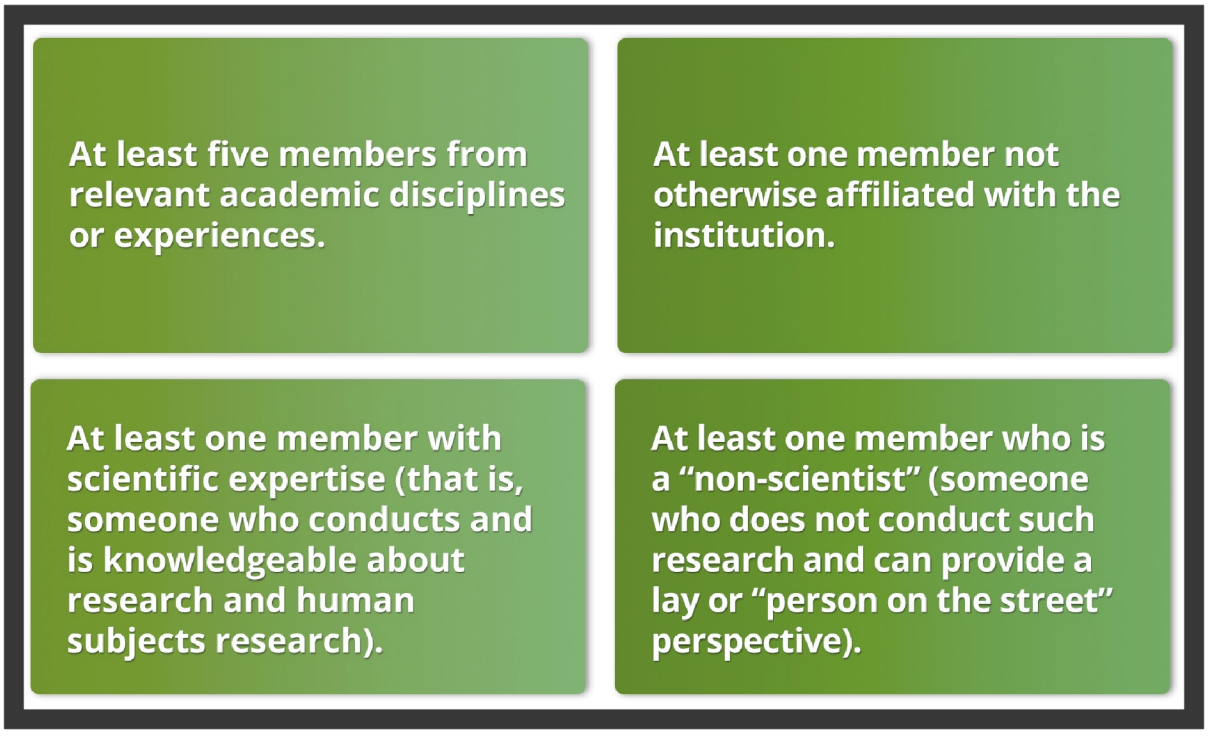
What is an Institutional Review Board?

Whether serving as the primary researcher or a research assistant, you must follow all institutional policies and procedures, best ethical practices, and applicable federal regulations. The IRB helps you meet those standards by providing expertise and resources, in addition to review and approval processes.



IRBs review and oversee human subjects research conducted by students, faculty, and staff. In this role, the IRB functions as a type of advocate to help protect the rights and welfare of human subjects by ensuring equitable subject selection, appropriate informed consent, risk mitigation, and respect for privacy and confidentiality. Depending on institutional policies, the IRB also may consider concerns for the rights and safety of researchers, although these types of considerations will most often be suggestions, not revision requirements.

IRB members must have the necessary experience and expertise to evaluate proposed research projects. The committee – which may include faculty, staff, students, and local community members – must be diverse in terms of race, gender, cultural backgrounds, and awareness of local community concerns. The IRB must include (45 CFR 46):



At larger institutions, the IRB may be part of a comprehensive, integrated system, such as the Human Research Protection Program (HRPP). The HRPP may include an Office for the Protection of Research Subjects, Office of Compliance, Office of Contracts and Grants, and other institutional officials and committees (for example, a community outreach program, biosafety, or radiation safety).

At smaller institutions, the IRB typically has a limited staff who handle all concerns regarding human research protections and report directly to one of the top academic officials.

Regardless of the structure and nature of the IRB or HRPP at your institution, staff members can address questions or concerns about the IRB review and submission process.

Types of IRB Review

Research projects that meet the regulatory definitions of “human subject” and “research” must receive IRB review. IRBs typically process research study plans under one of three review levels:



This section provides an overview to help you better understand the levels, but in most cases, the IRB will make the final determination.

In all cases, you must obtain written IRB approval (or notification of exemption) before initiating any aspect of a research study, including recruitment. IRBs cannot provide retroactive (after the fact) approval for any data previously collected for a study.

If you engage in human subjects research activities without IRB approval, your graduation or completion of your degree program could be delayed. You might be required to destroy the data or turn the data over to the IRB. Severe cases of

noncompliance (for example, failing to follow IRB or other institutional research policies) may result in a formal report of research misconduct.

Review for Exemption

Studies that involve no more than minimal risk and fit into one or more of the regulatory exempt categories are "exempt" from the Common Rule provisions and do not need further or ongoing review, unless the nature of the study changes so that it increases risk level. Review some examples of exempt studies.

If your project seems to fit one of the regulatory exempt categories, the IRB will then consider the following questions to assess the study's potential risk level:



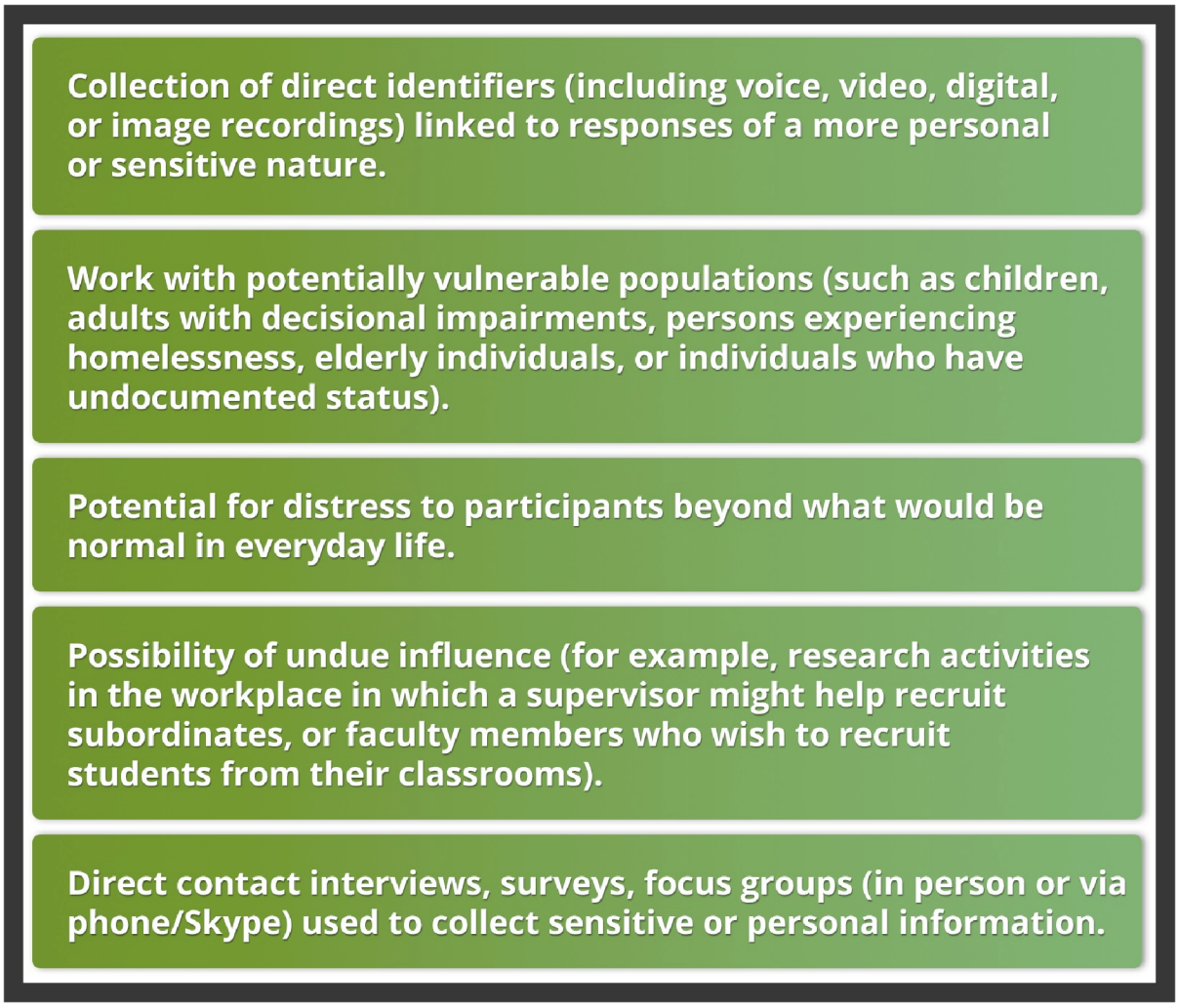
If the answer to any of these questions is yes, the IRB will likely require an expedited or convened committee review because the research may involve more than minimal risk of harm to subjects. Consult with your IRB to be sure.

The institution policies and procedures should identify who is able to review and make exempt determinations. This task is generally done by an IRB member or experienced staff person, and managed through the IRB office. Federal guidance

recommends that researchers do not make their own exempt determinations (OHRP 2020).

Expedited Review

Studies that require expedited review typically involve some degree of risk (for example, an increased privacy, psychological, or physical risk) to subjects, but still to a *minimal* degree. For expedited studies, a limited group of IRB members (the chair, members, and/or regulatory staff) will evaluate the study plan (instead of the convened committee). The federal regulations identify nine expedited categories (OHRP 1998). Some common characteristics of research reviewed using expedited review procedures include:



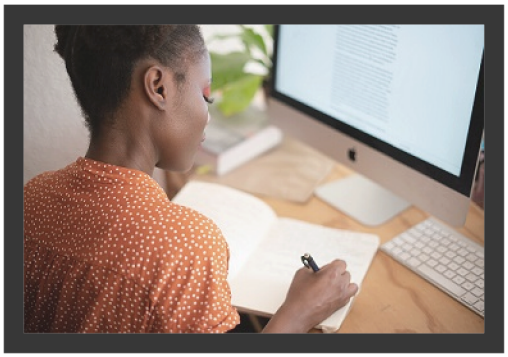
Convened Review

When research activities pose significantly higher risks to subjects than those that might occur in daily life, the project will need convened review (also commonly referred to as “full committee” or “full board” review) and must receive majority approval from members at the meeting. While student projects rarely involve this higher risk level, collecting data from certain populations or on certain topics might trigger convened review. Examples include research involving:

Minors (under the age of 18), if the data are sensitive	Pregnant women, fetuses, or newborns
Prisoners or persons under a correctional supervision (such as parolees)	Individuals with decisional impairment
Abuse, drug use, or sexuality	Certain biomedical and/or experimental procedures
Heightened physical discomfort, risk of injury, or invasion of privacy	Use of a regulated drug / device, dietary manipulation, or medical / clinical assessment of identifiable patients / volunteers

IRB Process for Student Researchers

Some institutions allow students to serve as lead researchers, while others require a faculty or staff member to serve as the primary or principal investigator, with the student listed as a co-investigator, secondary investigator, or research assistant. Regardless of your role, you should



assume responsibility for the process, pay close attention to detail, and compile the most professional study plan submission possible.

After you consult with the IRB and/or your faculty advisor to determine the correct review level and submission form(s) for your project, you should assemble the required documents for IRB review. At a minimum, most IRBs expect the following documents as part of a complete study plan.

- An application for IRB review that includes the plan for recruiting subjects, conducting the study, managing the data, and managing potential risks of harm to the subjects, and lists who will be involved. Some organizations may have different applications depending on whether the study might qualify for exemption.
- Recruitment materials such as flyers, email or telephone scripts, and social media announcements.
- Informed consent form or verbal script if you will collect oral consent.
- “Instruments” such as interview or focus group questions, surveys, or other procedures. While some IRBs require the exact questions you intend to ask, others may allow guiding questions or themes. Such decisions often relate to the sensitivity level of the topic or the research design.
- Human subjects training certification for all researchers.

Based on institutional or IRB policy, the IRB may also ask for other documents as a part of the submission, including:

- Site permission or approval (or approval from another IRB if one oversees the proposed research site)
- Approval from an oversight agency (for example, the U.S. Food and Drug Administration [FDA] if the study involves an FDA-regulated product or a state department of health if it is administratively responsible for oversight of a data repository; or approval from school boards or officials for permission to use school district data).

- Copyright or use permission (for example, if the study involves the use of a copyrighted instrument)

Work closely with your faculty advisor to ensure the submission is consistent and includes all required elements. Incomplete or inaccurate submissions will delay the review process. If your institution uses electronic submission software, it should guide you in providing all necessary documentation for a compete submission. The IRB office's website may list approximate IRB review timelines; if not, you or your faculty advisor can contact the IRB to inquire when you might receive feedback. If the IRB provides you with feedback, be sure to discuss all revisions with your faculty advisor before resubmission. If you do not understand a revision request or reviewer comment, email or call the IRB for further clarification.

Role of the Faculty Advisor



You may be able to choose your faculty advisor or the department/college may assign one to you. If you are able to choose, talk with other students about their experiences, so that you select someone who is knowledgeable about your field of study, supportive, and involved. Your faculty advisor will not complete your IRB submission but should provide guidance first and foremost in in developing a study plan that reflects best practices in the ethical and responsible conduct of human subjects research. The advisor should also ensure that you carry out these practices during the research activities. For instance, advisors should:

- Ensure you fully understand the importance of the informed consent process and discuss any necessary additional protections, depending on your intended subject population.
- Help you develop a secure plan for data storage in compliance with your institution’s data policies and best practices for securing potentially sensitive data.

- Verify your institution's data ownership policies to determine whether they must retain the raw data from the study or you must transfer any identifiable data to your faculty advisor prior to graduation.
- Make sure you secure written permission from institutions, schools, and/or other sites where research may take place.
- Review initial drafts of your study plan, as well as the final submission to check for completeness, accuracy, and best practices in human subject protections.
- Monitor your project as it progresses and ensure you maintain confidentiality, minimize risks and ensure voluntary participation and withdrawal.
- Help you report problems or negative events to the IRB right away, in compliance with institutional policy and federal regulations.

Ideally, your faculty advisor will provide clear guidance and support from beginning to end. Unfortunately, this standard of advising does not always occur. If you have difficulty communicating with or reaching your faculty advisor, or you feel you are not receiving correct information about the IRB process or human subjects research, contact the IRB for assistance and ask for a consultation with a staff member.

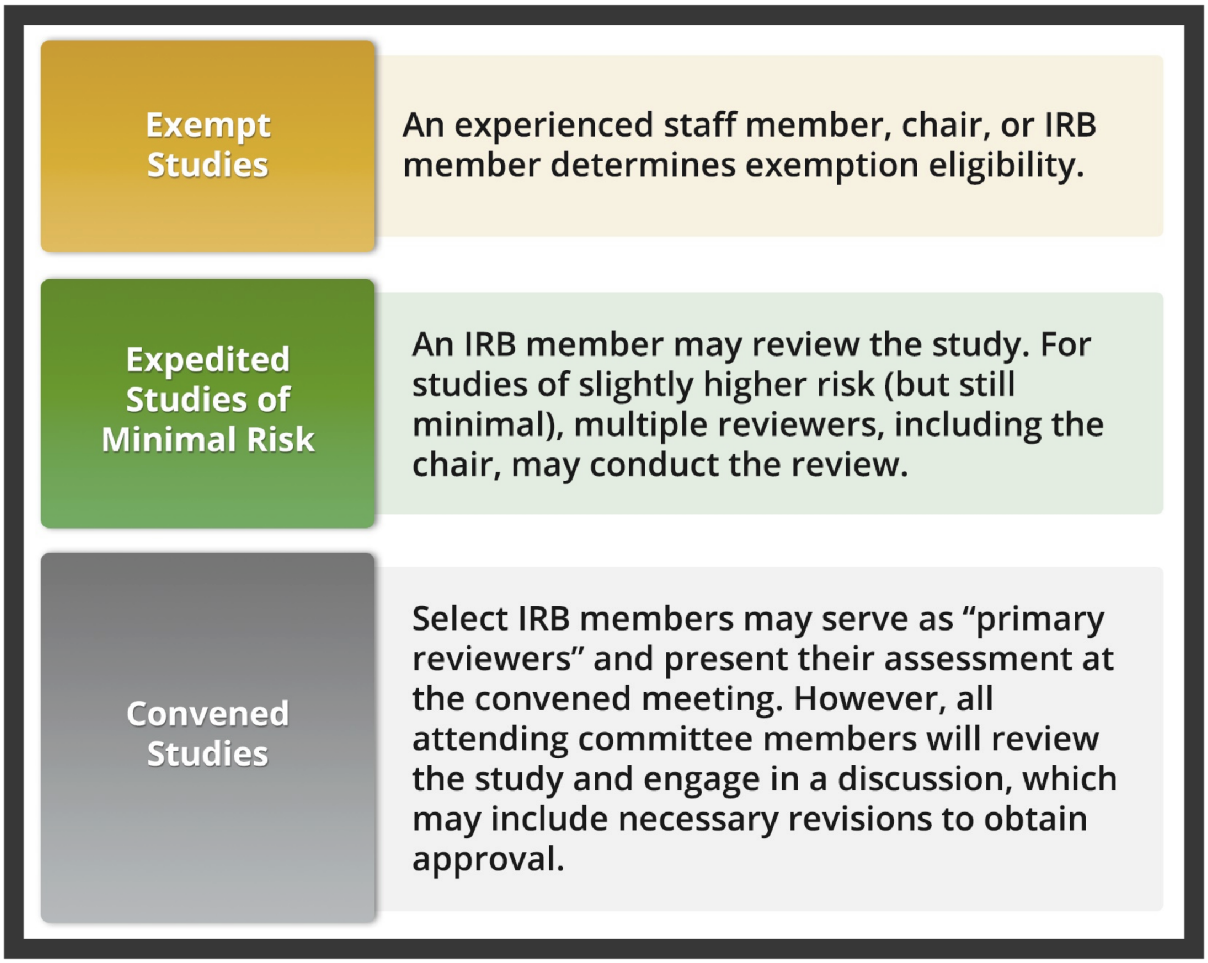
Department or Faculty Review and Approval

Some institutions involve another layer of review prior to the IRB process. Check with your department or college to find out whether a department chair, dean, faculty committee, scientific review committee, or faculty advisor must review and sign off on your application before submission to the IRB. This initial review may address scientific merit issues (that is, whether this research activity is worth completing, given your field), availability of resources, adherence to departmental policies, or other departmental issues. Find out the timeline as soon as possible for these preliminary review requirements because they may impact the larger timeline for conducting your research.

IRB Review

Check your IRB's website for deadlines and approximate timelines for review. You must allow enough time to obtain IRB approval and initiate your project. IRBs cannot "speed up" your review process nor can they ever provide retroactive approval if you collect data prior to IRB review and approval.

At many institutions, experienced IRB staff may initially screen your submission to ensure it is complete and ready for further review. Some staff may also determine exemption status or serve as expedited reviewers. Although IRBs may conduct the review process in multiple ways, common approaches are:



As mentioned previously, IRBs exist to review and mitigate risk to subjects and ensure equitable and beneficent research practices. IRBs should not request revisions to methodology unless the submission is unclear or incomplete or it poses a significant risk to subjects. Nor should IRBs review the main study plan application for items like grammatical or spelling mistakes.

On the other hand, the quality of your submission materials tells the IRB a great deal about you. If you have missing pieces, inconsistencies, typos, and grammatical and spelling errors in your protocol, what kind of researcher will you be in the field? You want to give the IRB confidence about your professionalism and capabilities based on a well-written, accurate submission.

In addition, typos and errors in your recruitment materials, informed consent form, and research instruments may steer potential subjects away from your study or make them question your credibility as a researcher. For this reason, and because your status as a student reflects on the reputation of the institution, IRBs will often ask you to revise any materials to be seen by potential subjects for accuracy and clarity.

IRB Review Outcomes

After the review process, the IRB may require revisions to help mitigate risk and/or ensure accuracy in the study plan. Responsibility for completing necessary revisions lies with the researcher, assisted by the faculty advisor. Once the IRB determines that all necessary revisions are complete, it will issue written approval and you may proceed with the study.

Occasionally, an IRB will defer (or "table" which means review at a later time) a study that shows high risk to subjects, unclear procedures, serious omissions, or ethical issues. In these cases, the IRB will notify you of concerns to address. In the case of a submission reviewed at a convened meeting, the IRB will reconsider the study plan at its next convened meeting.

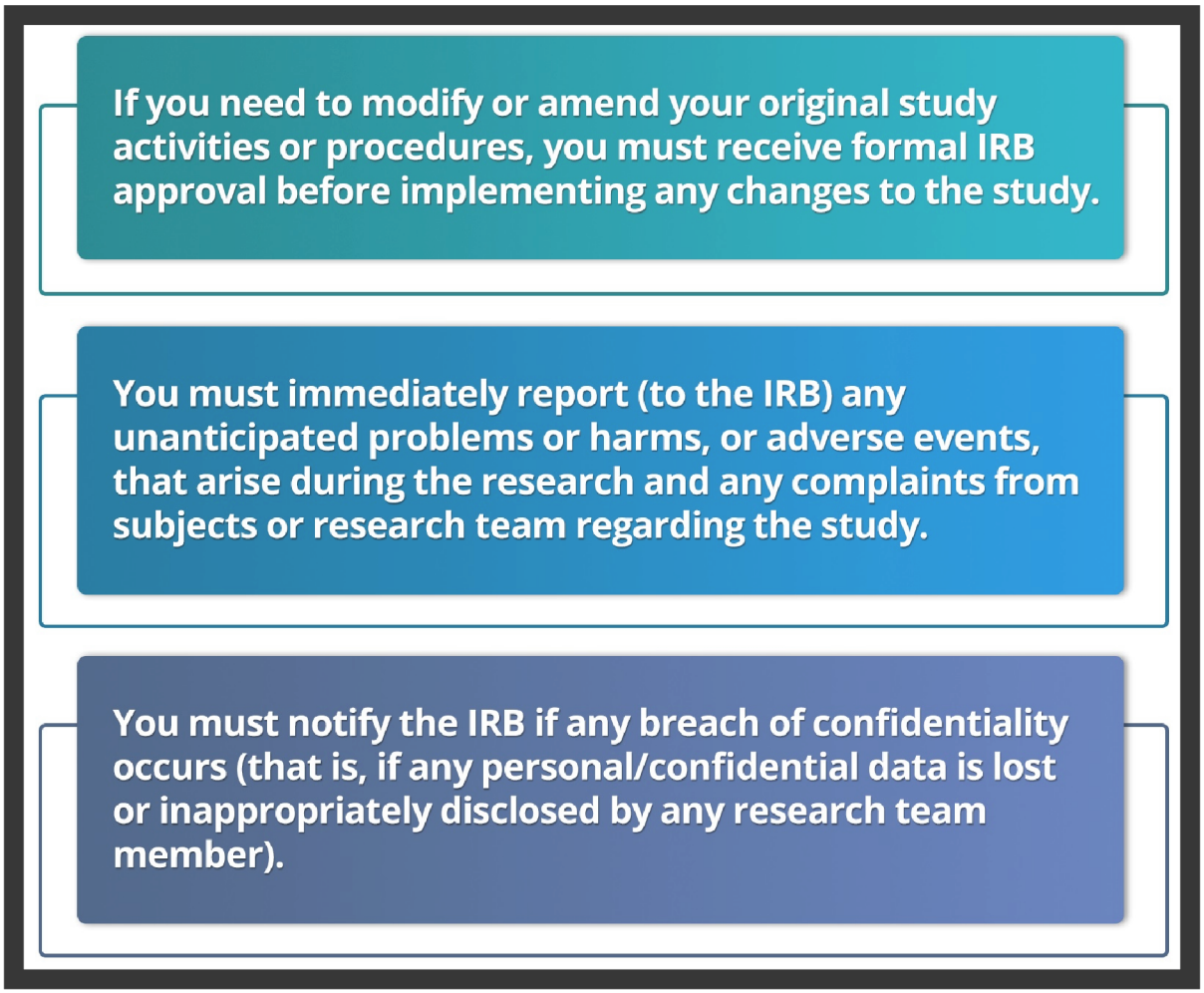
In rare circumstances, the IRB will determine that risks to subjects outweigh potential benefits to subjects and/or society. The IRB will "disapprove" the study plan, which means you may not conduct the study as presented (officials at the institution may not override IRB disapproval) (45 CFR 46).

It is likely that your study plan will only need minor revisions to meet approval criteria if you adequately work with your faculty advisor and consult with the IRB prior to

submission.

Post-Approval Research, Reporting, and Status Reports

When you receive written IRB approval, you may begin recruiting subjects and conducting study procedures. Certain situations require further interaction with the IRB:



Institutional requirements vary regarding status updates or closeout reports. Some institutions set student approval expirations in line with graduation dates, while others may establish an annual review. Closely read your approval letter for the applicable requirements, work closely with your faculty advisor throughout each stage, and consult with the IRB whenever questions arise.

Best Practices in Research Studies

Although institutional requirements and policies may vary, the following section reviews key elements of a study plan, as well as best practices to avoid pitfalls in the review process.

Recruitment

The following examples and considerations discuss different subject recruitment strategies.

Friends and Family

You may not foresee possible harms when recruiting from a population of your family and friends. However, when recruiting from known acquaintances, consider whether the nature of your study might lead people to share private information about other individuals without their permission.

For example, what if you ask your aunt whether your cousin might want to participate in a study about individuals who identify as non-binary? In this case, you may have unintentionally shared personal information about your cousin without their permission.

As another example, what if you post a link on social media to an online survey about legal recreational marijuana use and several of your friends tag other friends in the comments? Thereby revealing possible underage drug use or individuals' private activities that they may not want known publicly.

Snowball Sampling

With the snowball sampling method, others assist you in the recruiting process but should rarely provide you with potential subject names or contact information (for the reasons mentioned earlier about disclosing personal information without permission). Instead, the best practice for snowball sampling is to provide a link, an email, or a flyer that others can distribute to potential subjects, who may then choose

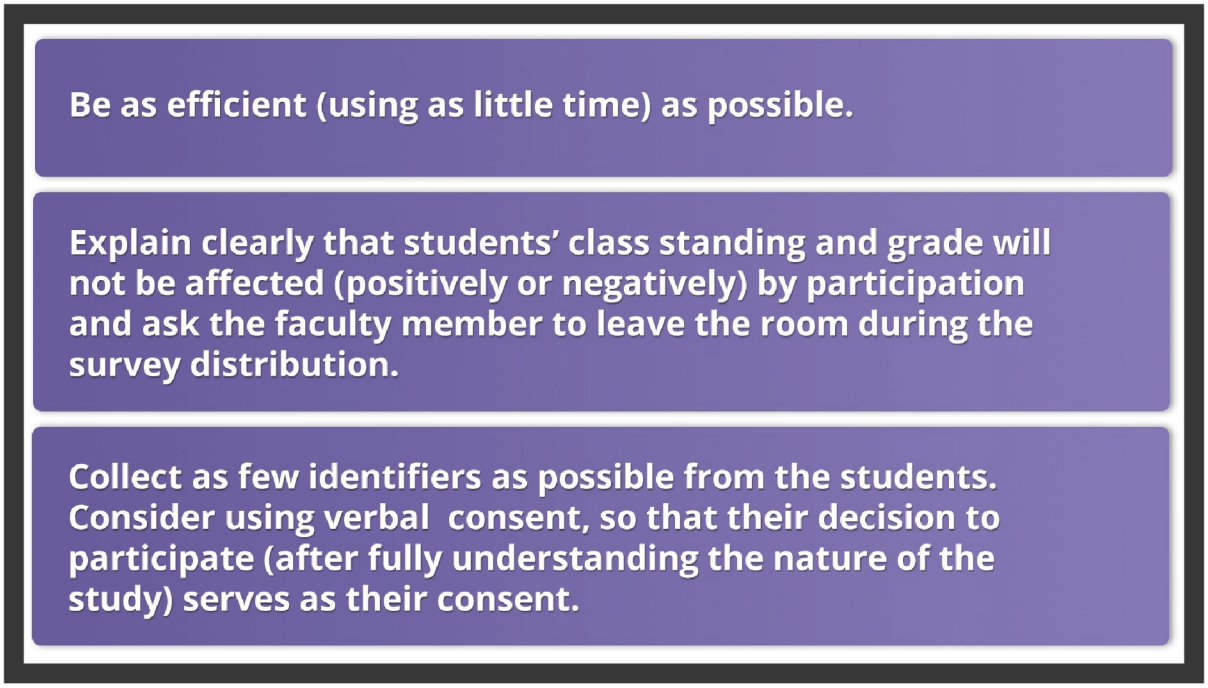
to click through (if a link) or contact you about participation. In this case, the series of events may reveal activities that certain individuals would rather keep private.

Social Media

In addition to using discretion in providing links to studies, you should also consider appropriate steps in collecting data from social media platforms. Generally, if someone expects their posts or photos to be private, you may not use that information for a study. Similarly, if you must register or obtain permission to join a group, then you may not gather data about that group without first disclosing your identity as a researcher and giving people the option to participate.

Classroom Populations

You might consider asking a current professor or other faculty member to distribute a survey in the classroom. If a faculty member agrees to let you distribute a very short survey or announce a link to a study or other study opportunity, you should take the following precautions:



Subject Pools

Your institution may have student subject pools that you can access for your research project. Often, students enrolled in introductory courses (such as introductory psychology) sign up within their department or school to participate in research projects and learn about the process as it relates to their field of study. The department or school will have its own guidelines for whether and how you may access these populations. You still need IRB approval, even if you recruit from an established subject pool.

Undue Influence in General

Always consider hierarchies or influences when thinking through recruitment.

For example, if you wish to interview staff at a middle school about whether they feel personally supported by their school administration, do you have a way to contact them aside from their school email addresses (which usually are not private, so the district may access employee email)? Will you meet with them at a location where their principal (or another supervisor) cannot see who participates?

Whether positive (individuals wish to participate out of belief that it will gain favor with a superior) or negative (individuals fear repercussions from a superior for participating), institutional hierarchies can affect how potential subjects feel about volunteering for a study.

Incentives/Compensation

IRBs must approve any incentives or compensation to subjects during their review of the study plan. Generally, IRBs discourage incentives for minimal risk studies, especially those conducted by students (however, snacks or food served at a focus group does not usually count as an incentive).

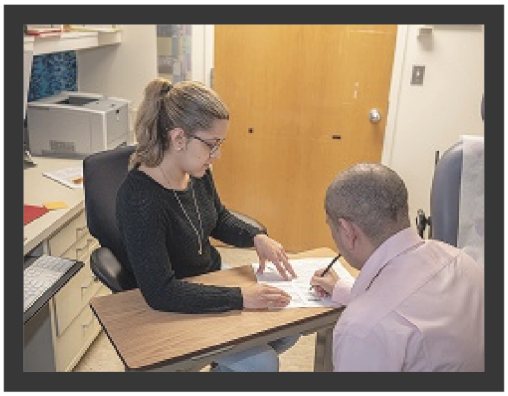
If you feel that a small gift for participating would be appropriate, be sure to consider the population. For international studies, even items like t-shirts or pens may create disharmony in small communities if members believe someone has been shown

higher favor. Similarly, clients at a food bank may feel financial pressure to participate in a survey if they will receive a $25 gift card for their time. These matters depend on local context, so if you are not familiar with a certain population, seek expert guidance to learn about culture and appropriate practices. Always check with your IRB to find out what institutional policies may affect incentives, related documentation, and recordkeeping.

Flyers, Emails, and Social Media Links

The informed consent process (as discussed in the next section) begins with clear, appropriate recruitment materials, so you should always include your study's intent and your contact information. An institution may also require other information pertinent to subjects on the recruitment material, including inclusion/exclusion criteria. Your institution may also require that you put its name or logo on the recruitment materials and/or indicate the flyers, emails, and other materials are "IRB approved." Check with your IRB to determine any institutional requirements.

Informed Consent Process



Informed consent is not a "one and done" process, you should view informed consent as an ongoing process throughout the study. Individuals should always be welcome to ask further questions, reconsider participation, retract certain data, or withdraw their data after the study takes place (realistically, however, if someone completes an online survey that collects no contact information or direct identifiers, you may be unable to screen out one individual's data.

Your consent materials must be clear and understandable to potential subjects, which means you should use non-technical, straightforward language with vocabulary around the sixth to eighth grade reading level. Specific regulations (45 CFR 46) require that you present key information to facilitate understanding and ensure consent

includes sufficient detail to assist an individual with making a decision regarding participation.

If necessary, translate consent documents into the intended population's language or include an interpreter on your research team (there may be specific regulatory and institutional requirements when recruiting a non-English speaking population, so consult with your IRB for guidance).

A basic informed consent form will include the following information (as required by 45 CFR 46) and any other specific institutional requirements.

- Contact information for you, your faculty advisor, and your IRB or compliance office.
- Information explaining that the subject is being asked to participate in research, including the purpose of the study, how long the study will last, and a detailed description of the tasks involved (including any interventions).
- Risks or discomforts the participant may experience.
- Potential benefits to subjects or others.
- The degree of confidentiality a subject can reasonably expect, which includes who will have access to the data, where and how you will store data, how you will report data/results (for example, in aggregate or using pseudonyms), and how you will use results (such as in reports, publications, or presentations).
- A statement that the individual's participation is voluntary, refusal to participate will not negatively impact the individual, and the individual may stop the study at any time without consequences.
- Any appropriate alternative treatments that may be advantageous to the subject instead of the research.
- For research greater than minimal risk, an explanation about available compensation and/or medical treatments if injury occurs, as well as where participants may obtain further information.
- For research that involves the collection of identifiable private information or biospecimens, a statement about possible future research studies and sharing

information or biospecimens with other researchers.

Note: The regulations may require additional consent elements, as applicable.

Alternatives to Standard, Written Consent

For certain minimal-risk studies, your IRB may approve or even suggest alternatives to standard, written consent, provided the alteration will not impact the subjects' rights. The following circumstances may merit an alternative to a signed consent document:



When using non-location surveys (such as mailed, emailed, or online), you must still provide a brief overview of the study's purpose, risks, and benefits, as well as your contact information. In the case of a non-location survey, returning or completing the survey represents that the subject consented and is an adult in that jurisdiction.

For certain populations, obtaining written consent itself may pose a risk. For example, these populations may include indigenous peoples who do not have or who may mistrust written language or people who wish to preserve anonymity of data (such as individuals who have undocumented status). In these cases, after you explain the study in language appropriate for the population, you should indicate in your notes that the subject understands the study and has given oral consent to participate. If you will conduct audio or video recording, you should record the oral consent.

For minimal risk studies on benign topics or for studies that collect no direct identifiers or too few indirect identifiers to determine an identity, you may request to

give potential subjects a study "information sheet" (including standard consent information). In these cases, an individual's decision to participate represents consent.

Typically, individuals provide consent. However, for research studies with minors (under the legal age to consent and generally 18 years old in most of the U.S.), you usually must obtain parental permission. Minors provide "assent" rather than "consent." You should write the assent materials at an appropriate language level, depending on the subjects' age. Additionally, you should check with your IRB regarding required parental signatures.

In some cases, an individual will have a legally authorized representative (LAR) who may provide consent on the individual's behalf. If you encounter this scenario, your faculty advisor and IRB will help you establish appropriate steps.

Anonymity, Confidentiality, and Privacy

This section explains the difference between anonymity, confidentiality, and privacy, so that you can learn how to protect subjects' private information.

> A good rule to follow is "Never gather more data than you truly need to answer your research question(s)."

Two important concepts lie at the heart of this discussion:

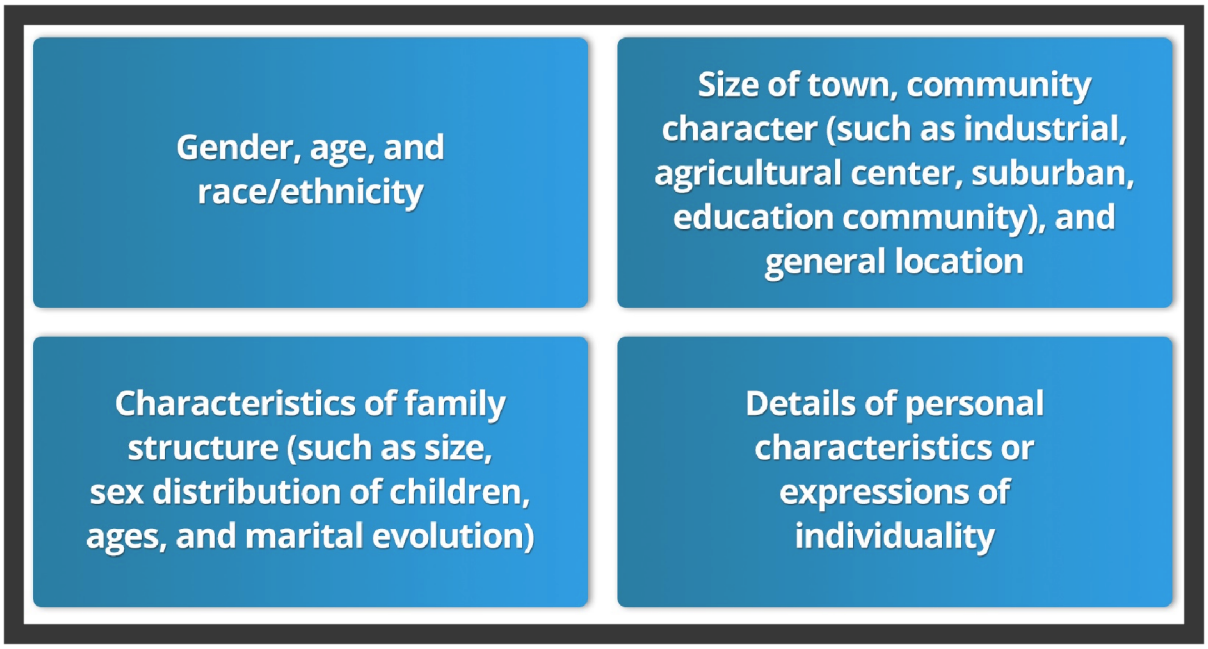


Direct identifiers may directly identify an individual. You should only collect them when necessary, particularly in connection with personal information. HIPAA's 18

personal identifiers are “protected” and can be seen as direct identifiers.

> The Health Insurance Portability and Accountability Act (HIPAA) of 1996 is a federal law that generally prohibits healthcare providers from disclosing protected health information (PHI) without written consent from the patient. Under HIPAA, PHI includes 18 personal identifiers such as an individual’s name, address, phone number, email address, and driver’s license number.

Indirect identifiers do not directly identify the subjects. However, even when you do not collect direct identifiers, a combination of other data might reveal an individual’s identity, especially with small sample sizes. These data include:



Anonymity

If the data are completely de-identified, they may be **anonymous** – meaning you do not know the identity of the respondent in connection with the data.

> **Caveats to Understanding Different Identifiability Levels of Data**
>
> - Interviews and video recordings that you conduct in person can never be “anonymous.”
> - Collecting any direct identifiers (such as name, address, phone number) that link responses to a specific individual or recording a list of codes associated with identifiers (such as names) means those data are not anonymous.

- Focus groups are never anonymous. If a combination of indirect identifiers (such as gender, race, and age) could identify a specific individual, particularly within study data from small samples, the study is not anonymous. Online surveys (such as those done using Qualtrics) can allow for anonymous data collection, but you must disable the feature collecting IP (internet protocol) addresses, and ensure that you do not collect indirect identifiers (such as gender, age, race, work-site) that identify specific individuals, particularly with small, unique populations.

Confidentiality

Confidentiality is an agreement (via informed consent) that potential subjects' individual responses and identities (their privacy) will not be disclosed beyond the research team without explicit permission.

You can rarely guarantee absolute confidentiality and must inform subjects of this. For example, you cannot control whether members of a focus group share others' information. In the case of a subject complaint, the IRB may need to review data and possibly consult with appropriate institutional officials (such as legal counsel). In addition, most institutions require that your faculty advisor has access to raw data, which the informed consent materials should mention.

Mandatory reporting laws may also impact confidentiality. If someone voices the intent to harm them self or others, you must report this intent to authorities. In colleges and universities, your faculty advisor is also a mandatory reporter in cases of Title IX violations (for example, discrimination based on sex, instances of sexual misconduct, or abuse).

Transmitting data via the internet (such as online or email surveys or videoconferencing interviews) always runs a risk of a confidentiality breach (that is, identifiable information released publicly without the participant's permission). You can minimize this risk by encrypting data transmission.

No time limit exists on the confidentiality of subject data. If you disseminate your results (such as in a research article, report, or presentation), you cannot identify subjects without their explicit permission. You may also not use photos, direct quotes, or descriptive information that could reveal an individual's identity if you do not have permission to do so.

Privacy

In research, you must protect an individual's **privacy**, which is the degree to which a subject has control over sharing personal information (such as physical, behavioral, or intellectual). The context includes timing, extent, and environment, during data collection and recruitment.

You must have permission to collect data (such as words or behavioral observations) for any situation where an individual has a reasonable expectation that no observation or recording will occur. An individual would not have a reasonable expectation of privacy if engaging in loud conversations or actions in places such as public transportation or the grocery store. However, in a classroom, church service, or closed online group, a person expects privacy and would not expect anyone to record notes for research purposes.

General Data Protection Regulation (GDPR) (EU 2016)

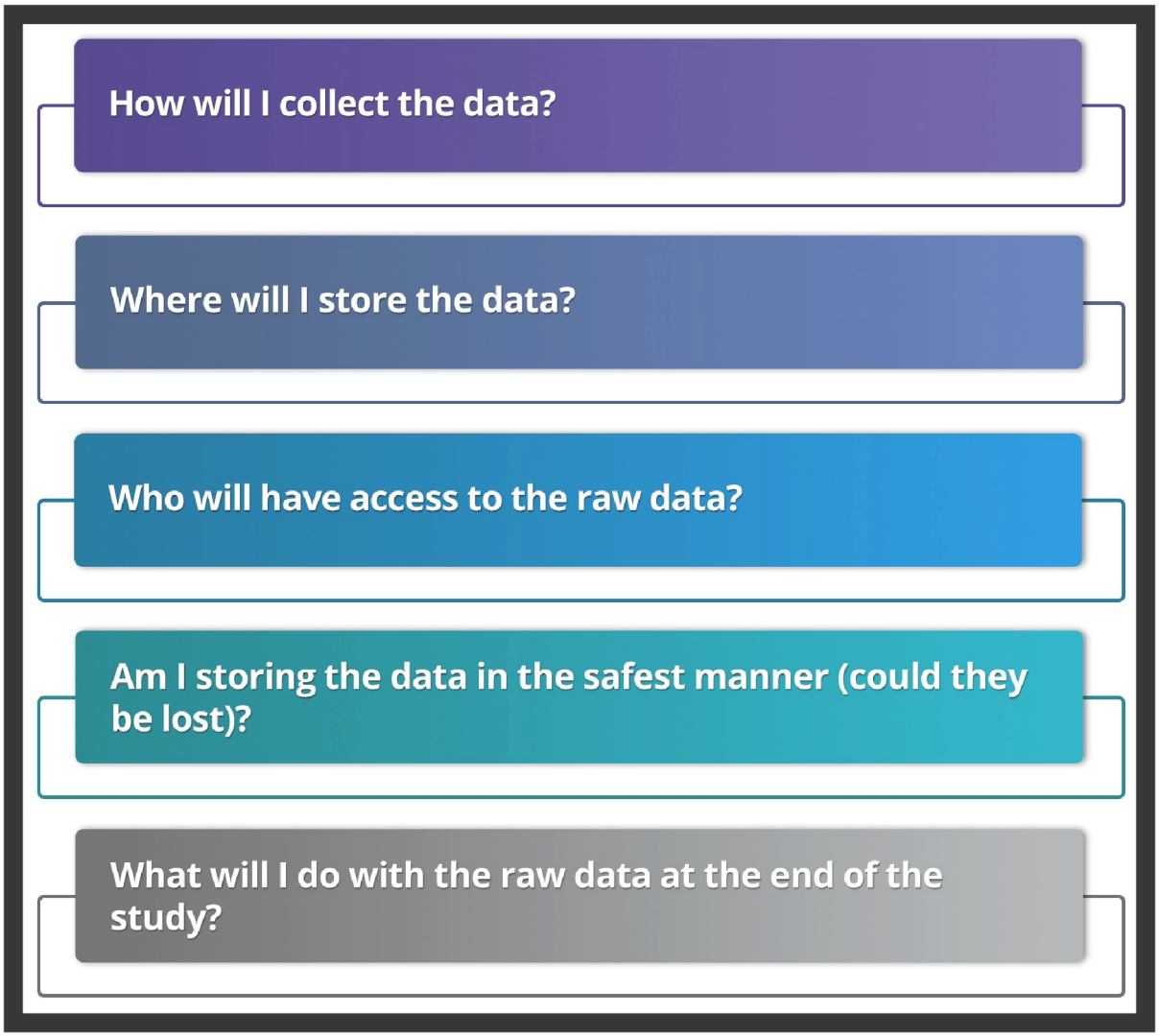
The GDPR governs the collection and use of personal information from all citizens of the European Union. In short, EU citizens have greater expectations of privacy and agency over what happens to their data after collection. You may not intend to collect data physically in the EU, but the GDPR even applies to online data collection. In other words, if you share a link via social media and any individuals in the EU somehow participate in your study, you are bound by this regulation (which carries significant penalties for noncompliance). At this stage, you may wish to avoid any possibility of EU subjects, unless you specifically need to recruit from that population. If so, work closely with your IRB to ensure full compliance with GDPR.

U.S. State and Local Laws

In the U.S., additional state and local laws may also exist to protect privacy.

Data Security and Storage Considerations

A best practice when handling subjects' data is to first review your institutional data privacy policy and any applicable IRB guidelines. You should then ask the following questions when designing the project:



Your faculty advisor should have access to any data collected as part of your research project. Institutional policies on data ownership vary, but all data should typically be fully deidentified *or* transferred to your faculty advisor's possession prior to graduation.

The following sections describe the data security and storage considerations for different types of data.

Paper, Audio, or Video Recordings

You can scan this type of data into an electronic format or store it in a secure location or keep it in a locked filing cabinet in a secure office.
Read more on paper, audio, or video recordings.

Electronic Data

You can secure electronic data by limiting access to authorized persons, as identified in your study plan and on the informed consent form.

Review some more considerations for storing electronic data.

Coded Data

To further protect an individual's data, you might wish to replace identifying information (such as names, addresses, or birthdates) with a code involving some combination of letters or numbers. You would then maintain a separate list with a code key, so you can use that key later to link identities and responses, if necessary.

If you wish to conduct longitudinal studies, where subjects complete multiple surveys, but you do not wish to collect direct identifiers, consider asking them to create an easily remembered code (such as the first three letters of a parent's or pet's name plus the last four digits of their cell phone number).

Non-Identifiable or Deidentified Data

IRBs consider data non-identifiable if:



You can de-identify data by removing all potential identifiers (direct or indirect) in a manner that prevents you or anyone else from re-identifying data in connection with an individual. Keep in mind that as technology records more of our personal lives, true deidentification and anonymity become more elusive.

Deception in Research

In some cases, you may wish to deliberately withhold information from subjects to collect data that is not influenced by participant knowledge of the research goal. Before approving "deception" methodologies, the IRB will assess whether:



Some deception studies will require convened IRB review due to potential risks to subjects, while others may meet exemption criteria due to a lack of significant risk.

International Research

Research involving human subjects outside of the U.S. may require additional protections to account for differences in language, culture, and social expectations. If you plan to conduct research outside of the U.S., you will need to provide additional information to the IRB to explain how you will address these differences.

Most submissions need prior approval from the appropriate authority, institution, or IRB equivalent in the country where you hope to collect data. If no equivalent ethics committee or group exists, you will likely need to demonstrate "local approval" from local experts or community leaders.

Federal regulations add further stipulations for conducting research outside of the U.S., including (45 CFR 46):

You must provide the same or equivalent protections to human subjects in research conducted in other countries.

You must know and comply with local laws, regulations, political and socio-economic factors, and cultural contexts for all locations where you wish to conduct research.

For higher risk studies, the IRB may request consultation with experts in the particular international setting.

The Office for Human Research Protections provides further resources on international research. You should also check with the education or study abroad office at your institution for any international travel policies that might require additional approval for student, faculty, or staff travel to high-risk areas (such as export control policies).

Reasonable Risks and Benefits

You must design and conduct studies that minimize potential risks of harm to subjects. In fact, IRBs will approve studies where the risks are minimized and reasonable in relation to anticipated benefits (45 CFR 46).

Risks of Harm



The Common Rule defines **minimal risk** as "the probability and magnitude of harm or discomfort anticipated in the research are not greater in and of themselves than those ordinarily encountered in daily life or during the performance of routine physical or psychological examinations or tests" (45 CFR 46) (Note – there is a slightly different definition of minimal risk for prisoners in the federal regulations).

The terms “ordinarily encountered” or “routine” may have broad interpretations. Thus, trained IRB members and staff assess study plans based on reasonable risk, which considers both the likelihood and magnitude of an event. For example, in a study that measures the maximum amount of oxygen athletes use during intense exercise, one risk of great magnitude could be cardiac arrest. Yet the likelihood of this event occurring among trained athletes who regularly engage in intense workouts is small. An IRB might approve this study, provided that the informed consent form clearly lays out all risks to subjects, and the researchers establish correct measures for appropriate monitoring.

Biomedical studies typically pose greater potential risks of physical harm to subjects, but social and behavioral studies also involve risks, often of a psychological, social, or economic nature.

For many student projects, participation may lead to only slight emotional or psychological risk. In these cases, your consent form should let potential subjects know that they could become upset or feel distress or embarrassment. In addition, the IRB may ask you to have a list of resources available to subjects who experience emotional distress.

Another risk is the possibility of a confidentiality breach. Especially in social and behavioral research, the disclosure of certain data could jeopardize a person’s reputation, social standing, or employment. On the other hand, you may need to breach confidentiality if you learn information about child or elder abuse or other illegal behavior. In some cases, such as with political or social activists, subjects may wish to disclose sensitive personal information, and even have their names and identities associated with the data, because they believe their role calls for such transparency.

If your informed consent form details potential risks of harm, including situations in which you might need to breach confidentiality, then your potential subjects can make the informed decision whether to take part in the study. The informed consent

form should represent a contract of confidentiality and ethical principles between you and the subjects.

Potential Benefits

Certain interventions may provide immediate or direct benefit to individual subjects. However, research studies often occur as part of a larger investigation within a field and may offer insights to that field or society. In these cases, your informed consent form should refer to reasonable benefits.

Students who intend to conduct a limited study with a small population often state in the study plan and on the consent form that the study will “fill a gap” in the field by “including marginalized voices” on a certain topic. While the study might indeed provide more insights from a small population, it is unlikely to fill any gaps in larger societal conversations.

Be realistic about what the study may achieve and limit your claims to reasonable expectations. Even if a study generates important information, it may take years for dissemination in the field or to create any measurable impact.

Remember that you should never list incentives and compensation, as well as food or other minor amenities, as benefits in the study plan or the informed consent form. Additionally, you should not consider it a benefit if subjects find participation rewarding or therapeutic or derive pleasure from helping.

Students as Research Subjects

In an academic setting, you might have the opportunity to both conduct research and participate as a research subject. For example, students often play an integral role in research that examines teaching methods, curricula, and other areas related to the scholarship of teaching and learning.



As you learned previously in this module, voluntary participation is a key principle of human subject protections, based on full and accurate information. This same principle applies to you as a research subject! If a faculty member wishes to recruit students for a research project, you should understand some best practices to ensure that the researcher respects your rights (and those of your fellow students). Ideally, the research project should relate to the topic of a class, and participation should be part of the learning experience for students.

Learn more about students as research subjects

If you or any other student feel you have been coerced to participate or continue in a study, or that your grade or class standing has been affected by your decision, you should talk to your institution's IRB or compliance office.

Student "Subject Pools"

As noted earlier, sometimes departments/schools create "subject pools," where students in introductory courses learn about research firsthand by signing up to participate in research projects. Researchers who wish the convenience of recruiting from these subject pools must submit their studies (usually exempt or expedited only) for IRB review and approval. Departments/schools may have certain expectations for what types of research may be conducted in this manner and specify who has access to participants.

Case Study

Students in Research

Gianna is a student in the Institute of Public Policy at Big State University. She is also considering a dual degree in sociology. Gianna

wants to study inclusive hiring practices at several large local companies in her area. She plans to interview the hiring managers about their companies' policies on inclusion in hiring.

- **Does her project require IRB review?**

Gianna receives the determination from the IRB that her project is not considered human subjects research, and she can proceed without IRB review and approval. But as she talks with hiring managers, she begins to wonder whether these companies actually reflect in practice their stated goals of inclusion and welcome-ness. She decides her research would benefit from including interviews with some employees willing to discuss their experiences in a work environment where they represent various minorities.

- **Does Gianna need to check back with the IRB before she conducts interviews with employees and explores a new research question?**
- **If Gianna doesn't collect names (a direct identifier) but does collect specific titles, as well as demographics such as gender, race, or aspects of physical ability, would this information be considered identifiable?**

Gina wants to collect data from employees on their experiences, including:

- General satisfaction in the workplace
- Personal experiences of sexism, racism, homophobia, transphobia, or discrimination based on ability
- Specific concerns or criticisms regarding the company, direct superiors, or colleagues

The more identifiers (direct or indirect) that Gianna wants to collect in addition to the degree of personal or revealing information, the greater the risk to the individual, who could face economic and social risks if their employment status is potentially affected by their responses.

Gianna submits her study protocol, and some of the IRB feedback questions include:

- How will the researcher ensure that participation remains confidential – in what location or environment will she conduct the interviews to maintain privacy?
- Will the researcher take handwritten notes or record the interviews in some manner? And how will she store the data securely?
- How long does an employee have after the interview to withdraw their data from the study, if they change their mind?

- **What additional safeguards should Gina consider including in her research to address these IRB concerns?**
- **Would the IRB be satisfied that there are appropriate protections in place?**

Close

Summary

This module provided information on the history and ethical principles of human subjects research and practical tips for navigating the IRB process.

You should now understand basic concepts, terminology, and strategies involved in developing and submitting a study plan to the IRB that offers appropriate protections to human subjects, in line with U.S. federal regulations and institutional policies.

Above all, contact your reviewing IRB at any stage of the process to seek guidance and assistance as you prepare to conduct your own human subjects research project.

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- *Trials of War Criminals before the Nuremberg Military Tribunals under Control Council Law No. 10 ("Green Series")*. Vol. 2. Washington D.C.: U.S. Government Printing Office (GPO), 1949b.

Additional Resources

- U.S. Department of Health and Human Services (HHS), Office for Human Research Protections (OHRP). 2003. “Guidance on Expedited Review Procedures.” Accessed March 15, 2016.

- U.S. Department of Health and Human Services (HHS), Office for Human Research Protections (OHRP). 2016. "Exempt Research Determination FAQs." Accessed August 18, 2016.

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